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Reverse Engineering ChatGPT Stories?

A Computer Science Perspective on Benzon's Story Experiments

Who is William Benzon?

Background

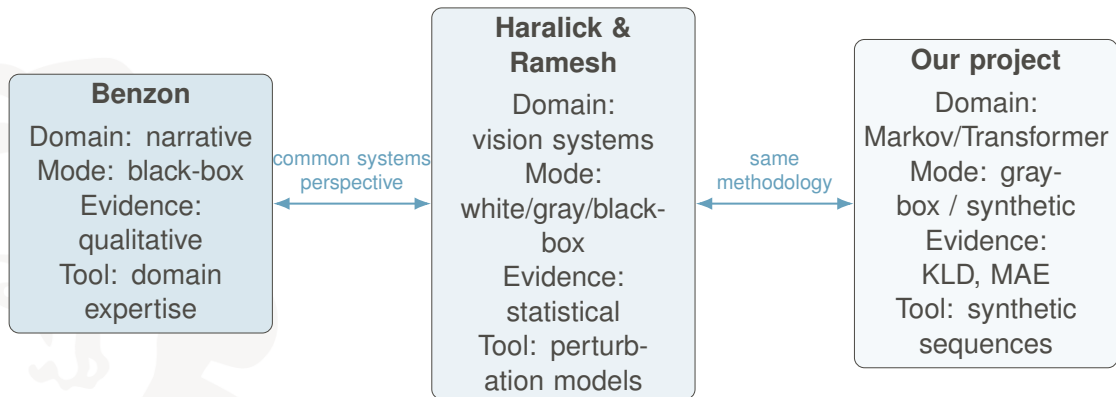
- ▶ PhD in English at SUNY Buffalo; worked in the intellectual tradition of David G. Hays with long-standing research in cognitive science and computational semantics.
- ▶ Joined Hays' computational linguistics group in 1974.
- ▶ Independent researcher: literary theory, cognitive science, AI.

His framework

- ▶ Cognitive networks: concepts as nodes, relations as edges.
- ▶ Abstract concepts defined over patterns in stories.
- ▶ Developed computational and cognitive theories of narrative over several decades.

Key point

Benzon is a **domain expert** with a long-standing background in cognitive science and computational semantics — not a naive literary critic.



Common abstract question

Perturb a system with a known input structure. Observe what is preserved. Characterise the system's behaviour.

Main question

What can Benzon's story experiments tell us about the behaviour of ChatGPT, viewed from a computer science perspective?

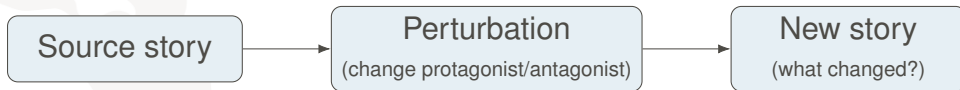
- ▶ Explain Benzon's method and main hypothesis.
- ▶ Translate narrative concepts into CS / ML terminology.
- ▶ Connect his black-box probing to the Haralick–Ramesh performance characterisation framework.
- ▶ Show how our Transformer experiments follow the same systems-level perspective.

Framing

Benzon and Ramesh are **complementary** — same abstract question, different domains, different levels of quantitative rigour.

Benzon's idea

Change one element in a ChatGPT story and observe how the rest changes — as a probe of the model's **learned behaviour**.



Example: Princess Aurora → Prince Harry / Henry the Eloquent / William the Lazy.

In systems terms

A controlled input perturbation is applied to a black-box system. The output is analysed to characterise the system's **behaviour under perturbation**.

- ▶ Start from a ChatGPT story about Princess Aurora.
- ▶ Ask ChatGPT to retell it with a changed protagonist or antagonist.
- ▶ Align old and new stories paragraph by paragraph.
- ▶ Highlight differences — interpreted as signs of narrative coherence.

Domain expertise as the measurement instrument

Benzon uses his background as a literary analyst and cognitive scientist to interpret what is preserved and what changes — **black-box probing mediated by domain knowledge**.

Methodological observation

It is not clearly specified whether all 12 experiments were conducted in one session or independently — this may influence reproducibility and interpretation.

Benzon divides stories into five segments — a parse-tree-like decomposition:

Segment	Function
Donné	Initial situation / background
Disturbance	Problem or threat appears
Plan / Transit	Hero reacts and moves toward solution
Enact	Main action against the problem
Celebrate	Resolution and social recognition

Connection to formal linguistics

These segments resemble a **story grammar** — a hierarchical decomposition into structural primitives. Do they reflect latent behavioural regularities of the model, or are they imposed by the analyst?

Example: Aurora becomes Harry

Aurora

- ▶ Kind princess
- ▶ Singing voice
- ▶ Calms dragon by singing
- ▶ Returns as hero

Harry

- ▶ Kind prince
- ▶ Sword-fighting skills
- ▶ Defeats dragon in battle
- ▶ Returns as hero

What is preserved / what changes

Global trajectory (Disturbance, Celebrate) is preserved. The protagonist's defining attribute and conflict resolution change. The perturbation propagates **locally**, not uniformly.

Example: Aurora becomes William the Lazy

The tension

William must be a hero *and* be lazy — two constraints in conflict.

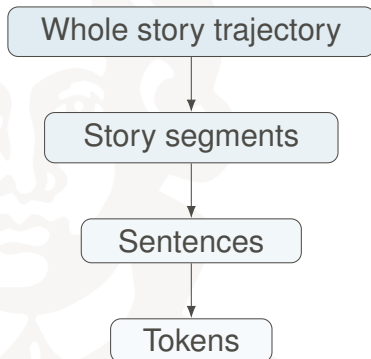
- ▶ Aurora charges the dragon and sings.
- ▶ William stays in his castle, concocts a potion, delegates to knights.
- ▶ Plan/Transit, Enact, and Celebrate all change substantially.
- ▶ Disturbance remains unchanged — independent of the protagonist.

Systems interpretation

The epithet “lazy” acts as a **parameterised constraint** shifting the generation behaviour. The degree of change appears proportional to the semantic distance from the original character.

Nested hierarchy of probability distributions

Benzon conjectures that ChatGPT's generation may be organised hierarchically across story, segment, sentence, and token levels.



Signal processing view

- ▶ A possible latent narrative organisation guides generation.
- ▶ Each level may constrain the behaviour below.
- ▶ What is preserved under perturbation is consistent with this hierarchy.

Benzon's term	CS / ML / systems interpretation
Story trajectory	High-level behavioural regularity of generated sequence
Segment	Local discourse state; node in a narrative parse tree
Protagonist / antagonist	Named agent / primitive in the story grammar
Transformation	Controlled input perturbation
Coherence	Consistency of generated continuation under structural constraints
Probability hierarchy	Context-dependent conditional distribution over next tokens
Reverse engineering	Black-box behavioural probing via systematic perturbation

Key distinction

Benzon observes output behaviour only — no access to weights, activations, or attention heads. Systematic, but black-box.



Domain expert interprets
what changed and why

- ▶ Benzon's domain expertise is the **measurement instrument**.
- ▶ He identifies which behavioural regularities are preserved and which change.
- ▶ Analogous to a vision expert who designs probes based on knowledge of what features matter.

Key question

What **behavioural regularities** does ChatGPT's training instil — what learned “recipe” does the model follow?

Core idea

Introduce **controlled perturbations** to a system with known input structure and measure how output performance degrades as a function of perturbation parameters.

Component	Role
Known input / domain model	Ground truth; enables fidelity measurement
Perturbation / operating conditions	Controlled input variation
Performance metric	Quantitative output characterisation
White/gray/black-box mode	Degree of access to system internals

Goal

Move from “it works on my examples” to a rigorous **engineering characterisation** of system behaviour and performance limits.

	Benzon	Haralick & Ramesh
System	ChatGPT (opaque LLM)	Vision algorithm
Known input	Assumed / interpreted narrative structure	Synthetic data from known distribution
Perturbation	Character substitution	Random noise / operating conditions
Measurement Mode	Domain expert interpretation Black-box	Statistical metrics White / gray / black-box
What is preserved?	Story trajectory, coherence	Algorithm output under degradation

Common systems perspective

Both ask: **what does the system preserve when the input is perturbed?**
The difference is the language and level of quantitative rigour.

Motivation

Are Benzon's observed transformations reproducible? Does session history affect results?

Characters tested:

- ▶ William the Lazy
- ▶ Henry the Eloquent
- ▶ Prince Harry

Conditions:

- ▶ Independent sessions
- ▶ Same-session transformations

Purpose

Preliminary qualitative replication before committing to a full quantitative protocol — consistent with a black-box probing approach.

Character	Main Adaptation	Observation
William the Lazy	Laziness integrated into actions	Consistent
Henry the Eloquent	Eloquence changes abilities and artifacts	Larger modifications
Prince Harry	Title and gender adaptation	Close to original

Qualitative observation

Across our small sample, larger semantic differences tended to produce larger narrative adaptations — consistent with Benzon's conjecture and the perturbation–performance relationship in Haralick & Ramesh.

Limitation

Preliminary qualitative replication — not statistical validation.

Segment	Stability	Observation
Donné	Low	Character traits adapted
Disturbance	High	Conflict episode preserved
Plan / Action	Medium	Trait-dependent modifications
Celebrate	Medium	Moral lesson adapted

Systems interpretation

The perturbation is **absorbed differently at each level of the hierarchy**. The Disturbance segment behaves like a stable part of the global trajectory; Donné and Enact are more sensitive — consistent with a hierarchically organised generative behaviour.

Related paper: 20 versions of a prototypical story

- ▶ Minimal prompt: `story` — no structure specified.
- ▶ 10 stories in one session; 10 in separate sessions.
- ▶ 19 out of 20 follow the same exploration-and-return archetype.
- ▶ Session history affects protagonist variety.

What this reveals

Repeated sampling under minimal conditions reveals the model's **preferred operating behaviour** — a learned narrative prior that emerges consistently without any structural prompt.

Methodological implication

Session context is an uncontrolled variable — a nuisance variable in Haralick–Ramesh terms that should be controlled in a formal protocol.

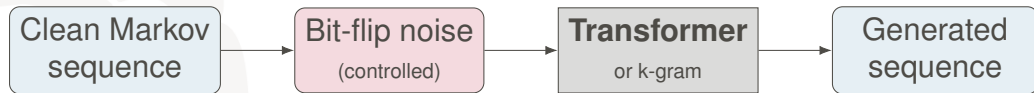
Inspired by the performance characterisation methodology of Haralick & Ramesh

Rather than analysing natural stories directly, we construct synthetic stochastic processes with known statistical properties and measure what the Transformer preserves.

	Benzon	Our project
Object	Stories	Binary Markov sequences
Known structure	Assumed narrative structure	Transition probabilities
Perturbation	Character substitution	Bit-flip noise / order / scale
Measurement	Domain expert interpretation	KLD, MAE, statistical tests
Ground truth	Unknown	Known stochastic process
Mode	Black-box	Gray-box / controlled synthetic

Setup

Transformers trained on clean Markov sequences; bit-flip noise added only at test time.



- ▶ k-gram: error increases with noise — sensitive to input.
- ▶ Transformer: error stays flat — behaviour dominated by learned weights.

What is preserved?

The Transformer preserves the training distribution even under corrupted input — generation from learned behaviour, not noisy context.

	Benzon	Our project
Perturbation	Change protagonist	Add bit-flip noise
What is preserved	Story trajectory, global coherence	Transition probabilities
What changes	Segment-level attributes	Little observed change under tested noise levels
Interpretation	Possible latent narrative organisation	Distribution approximated in learned behaviour

Shared finding

Both are consistent with the same phenomenon: model output behaviour is dominated by **learned organisation**, not by the specific input perturbation.

Common systems perspective

Although the domains differ — stories versus stochastic sequences — all three works investigate the same scientific question: how can we characterise the behaviour of a complex system by systematically perturbing its inputs and analysing the outputs?

How could Benzon's approach be made more rigorous?

Following the Haralick–Ramesh performance characterisation protocol:

- ▶ Generate n stories per condition — not one example.
- ▶ Control model version, temperature, top-p, and session history.
- ▶ Define m perturbation types: gender, role, personality trait, antagonist.
- ▶ Automatically detect story segments.
- ▶ Use embeddings to quantify protagonist distance.
- ▶ Test statistically whether perturbation magnitude predicts segment-level change.

From observation to characterisation

Move from “I observe a pattern” to “the generated distribution differs significantly from the reference under a defined metric.”

Takeaway

Benzon and Ramesh share a common systems perspective: perturb, observe what is preserved, characterise behaviour. Different domains, different rigour — complementary approaches.

- ▶ Benzon is a domain expert whose black-box probing is principled and valuable.
- ▶ His experiments **suggest** that ChatGPT may preserve narrative structure under perturbation.
- ▶ Our noise experiments are consistent with the same phenomenon quantitatively.
- ▶ The bridge is **transdisciplinary harmonisation** — translating narrative hypotheses into statistically testable claims using the language of distributions and systems theory.

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- Haralick, R.M. (1992). *Performance characterization in computer vision*. BMVC.
- Ramesh, V. and Haralick, R.M. (1991). *Performance characterization of vision algorithms*. Proc. SPIE 1406.
- Ramesh, V. and Dori, D. (2015). *A Model-Based Framework for Developing Certifiable Autonomous Cognitive Systems*. Technical report.